



Question 1

Reverse words order in a string

Input

code gemini

Output

gemini code

Answer

```
s = "code gemini"
result = " ".join(s.split()[::-1])
print(result)
```

Question 2

Reverse complete string

Input

code gemini

Output

inimeg edoc

Answer

```
s = "code gemini"  
print(s[::-1])
```

Question 3

Reverse each word but keep order same

Input

code gemini

Output

edoc inim eg

Answer

```
s = "code gemini"  
result = ".join(word[::-1] for word in s.split())  
print(result)
```

Question 4

Count frequency of each character

Input

genai

Answer

```
s = "genai"  
freq = {}  
  
for ch in s:  
    freq[ch] = freq.get(ch, 0) + 1  
  
print(freq)
```

Question 5

Find maximum repeated character

Input

programming

Output

g

Answer

```
s = "programming"
```

```
freq = {}
```

```
for ch in s
```

```
    freq[ch] = freq.get(ch, 0) + 1
```

```
max_char = max(freq, key=freq.get)
```

```
print(max_char)
```

Question 6

Find maximum repeated word

Input

ai gen ai agent ai gen

Answer

```
text = "ai gen ai agent ai gen"
```

```
words = text.split()
```

```
freq = {}
```

```
for w in words
```

```
    freq[w] = freq.get(w, 0) + 1
```

```
print(max(freq, key=freq.get))
```

Question 7

Check palindrome string

Input

madam

Answer

```
s = "madam"
print(s == s[::-1])
```

Question 8

Remove duplicate characters but keep order

Input

aabbccdd

Output

abcd

Answer

```
s = "aabbccdd"
result = ""
```

```
for ch in s:
    if ch not in result:
        result += ch
```

```
print(result)
```

Question 9

Check anagram strings

Input

listen silent

Answer

```
s1 = "listen"
```

```
s2 = "silent"
```

```
print(sorted(s1) == sorted(s2))
```

Question 10

First non repeating character

Input

swiss

Output

w

Answer

```
s = "swiss"
```

```
freq = {}
```

```
for ch in s
```

```
    freq[ch] = freq.get(ch, 0) + 1
```

```
for ch in s
```

```
    if freq[ch] == 1
```

```
        print(ch)
```

```
    break
```

Question 11

Remove special characters

Input

ai@gen#2024!

Output

aigen2024

Answer

```
s = "ai@gen#2024!"
```

```
result = ""
```

```
for ch in s
```

```
    if ch.isalnum():
```

```
        result += ch
```

```
print(result)
```

Question 12

Capitalize first letter of each word

Input

gen ai agent

Answer

```
s = "gen ai agent"
```

```
print(s.title())
```

Question 13

Find longest word in string

Input

gen ai autonomous agent system

Answer

```
s = "gen ai autonomous agent system"
words = s.split()
print(max(words, key=len))
```

Question 14

Count vowels and consonants

Answer

```
s = "generativeai"
vowels = "aeiou"
v_count = 0
c_count = 0
```

```
for ch in s
```

```
    if ch in vowels
```

```
        v_count += 1
```

```
    else
```

```
        c_count += 1
```

```
print(v_count, c_count)
```

Question 15

String compression

Input

aabbbccccc

Output

a2b3c4

Answer

```
s = "aabbcbccc"
```

```
result = ""
```

```
count = 1
```

```
for i in range(1, len(s))
```

```
    if s[i] == s[i-1]
```

```
        count += 1
```

```
    else
```

```
        result += s[i-1] + str(count)
```

```
        count = 1
```

```
result += s[-1] + str(count)
```

```
print(result)
```

Question 16

Generate all substrings

Input

abc

Answer

```
s = "abc"
```

```
substrings = []
```

```
for i in range(len(s))
```

```
    for j in range(i + 1, len(s) + 1)
```

```
        substrings.append(s[i:j])
```

```
print(substrings)
```

Question 17

Remove extra spaces

Input

gen ai agent

Answer

```
s = " gen ai agent "  
print(" ".join(s.split()))
```

Question 18

Replace word without using replace function

Input

ai is future

Replace ai with genai

Answer

```
s = "ai is future"  
words = s.split()  
  
for i in range(len(words))  
    if words[i] == "ai"  
        words[i] = "genai"  
  
print(" ".join(words))
```

Question 19

Check string contains only digits

Input

12345

Answer

```
s = "12345"  
print(s.isdigit())
```

Question 20

Separate alphabets and digits

Input

A1B2C3

Output

ABC123

Answer

```
s = "A1B2C3"  
letters = ""  
digits = ""  
  
for ch in s:  
    if ch.isalpha():  
        letters += ch  
    else:  
        digits += ch  
  
print(letters + digits)
```

Question 21

Split string and store words into list

Input

code gemini

Output

['code', 'gemini']

Answer

```
s = "code gemini"  
words = s.split()  
print(words)
```

Question 22

Split string and reverse list

Input

code gemini ai

Output

['ai', 'gemini', 'code']

Answer

```
s = "code gemini ai"  
words = s.split()  
words.reverse()  
print(words)
```

Question 23

Convert string into list of characters

Input

genai

Output

```
['g', 'e', 'n', 'a', 'i']
```

Answer

```
s = "genai"  
chars = list(s)  
print(chars)
```

Question 24

Count words in a sentence

Input

gen ai agent system

Output

4

Answer

```
s = "gen ai agent system"  
print(len(s.split()))
```

Question 25

Find word with maximum length

Input

code gemini artificial intelligence

Output

intelligence

Answer

```
s = "code gemini artificial intelligence"
```

```
words = s.split()
longest = words[0]

for w in words
    if len(w) > len(longest)
        longest = w

print(longest)
```

Question 26

Sum of digits in a string

Input

a1b2c3

Output

6

Answer

```
s = "a1b2c3"
total = 0

for ch in s
    if ch.isdigit()
        total += int(ch)

print(total)
```

Question 27

Extract numbers from string

Input

ai2024gen5

Output

20245

Answer

```
s = "ai2024gen5"
```

```
numbers = ""
```

```
for ch in s
```

```
    if ch.isdigit()
```

```
        numbers += ch
```

```
print(numbers)
```

Question 28

Check Armstrong number

Input

153

Output

True

Answer

```
n = 153
```

```
temp = n
```

```
total = 0
```

```
while temp > 0
```

```
    digit = temp % 10
```

```
total += digit ** 3
temp = temp // 10

print(total == n)
```

Question 29

Check Palindrome number

Input

121

Output

True

Answer

```
n = 121
```

```
temp = n
```

```
rev = 0
```

```
while temp > 0
```

```
    rev = rev * 10 + temp % 10
```

```
    temp = temp // 10
```

```
print(rev == n)
```

Question 30

Find factorial of a number

Input

5

Output

120

Answer

n = 5

fact = 1

for i in range(1, n + 1)

fact *= i

print(fact)

Question 31

Check Prime number

Input

13

Output

True

Answer

n = 13

is_prime = True

if n <= 1

is_prime = False

else

for i in range(2, n)

if n % i == 0

is_prime = False

break


```
print(is_prime)
```

Question 32

Generate Fibonacci series

Input

5

Output

0 1 1 2 3

Answer

```
n = 5
```

```
a = 0
```

```
b = 1
```

```
print(a)
```

```
print(b)
```

```
for i in range(n - 2)
```

```
    c = a + b
```

```
    print(c)
```

```
    a = b
```

```
    b = c
```

Question 33

Find largest number in list

Input

[4, 7, 2, 9, 1]

Output

9

Answer

```
nums = [4, 7, 2, 9, 1]
```

```
largest = nums[0]
```

```
for n in nums
```

```
    if n > largest
```

```
        largest = n
```

```
print(largest)
```

Question 34

Find second largest number

Input

```
[10, 5, 8, 20]
```

Output

10

Answer

```
nums = [10, 5, 8, 20]
```

```
first = second = -1
```

```
for n in nums
```

```
    if n > first
```

```
        second = first
```

```
        first = n
```

```
    elif n > second and n != first
```

```
        second = n
```

```
print(second)
```

Question 35

Remove duplicates from list

Input

```
[1, 2, 2, 3, 4, 4]
```

Output

```
[1, 2, 3, 4]
```

Answer

```
nums = [1, 2, 2, 3, 4, 4]
```

```
unique = []
```

```
for n in nums
```

```
    if n not in unique
```

```
        unique.append(n)
```

```
print(unique)
```

Question 36

Check list is sorted or not

Input

```
[1, 2, 3, 4]
```

Output

```
True
```

Answer

```
nums = [1, 2, 3, 4]
```

```
is_sorted = True
```

```
for i in range(len(nums) - 1)
```

```
    if nums[i] > nums[i + 1]
```

```
        is_sorted = False
```

```
        break
```

```
print(is_sorted)
```

Question 37

Find sum of list elements

Input

```
[1, 2, 3, 4]
```

Output

```
10
```

Answer

```
nums = [1, 2, 3, 4]
```

```
total = 0
```

```
for n in nums
```

```
    total += n
```

```
print(total)
```

Question 38

Count even and odd numbers

Input

[1, 2, 3, 4, 5]

Output

Even 2

Odd 3

Answer

```
nums = [1, 2, 3, 4, 5]
```

```
even = 0
```

```
odd = 0
```

```
for n in nums
```

```
    if n % 2 == 0
```

```
        even += 1
```

```
    else
```

```
        odd += 1
```

```
print(even, odd)
```

Question 39

Swap two numbers without third variable

Input

a = 5 b = 10

Answer

a = 5

b = 10

a, b = b, a

print(a, b)

Question 40

Count digits in a number

Input

12345

Output

5

Answer

```
n = 12345
```

```
count = 0
```

```
while n > 0
```

```
    count += 1
```

```
    n = n // 10
```

```
print(count)
```

Question 41

Print numbers from 1 to 100 without using loop

Answer

```
def print_numbers(n)
```

```
    if n > 100
```

```
        return
```

```
    print(n)
```

```
    print_numbers(n + 1)
```

```
print_numbers(1)
```

Question 42

Print even numbers between 1 and 50

Answer

```
for i in range(1, 51)
    if i % 2 == 0
        print(i)
```

Question 43

Find missing number in list

Input

[1, 2, 3, 5]

Output

4

Answer

```
nums = [1, 2, 3, 5]
n = len(nums) + 1
total = n * (n + 1) // 2
print(total - sum(nums))
```

Question 44

Move all zeros to end

Input

[1, 0, 3, 0, 5]

Output

[1, 3, 5, 0, 0]

Answer

```
nums = [1, 0, 3, 0, 5]
```

```
result = []
```

```
count = 0
```

```
for n in nums
```

```
    if n != 0
```

```
        result.append(n)
```

```
    else
```

```
        count += 1
```

```
result.extend([0] * count)
```

```
print(result)
```

Question 45

Find intersection of two lists

Input

[1, 2, 3] and [2, 3, 4]

Output

[2, 3]

Answer

```
a = [1, 2, 3]
```

```
b = [2, 3, 4]
```

```
common = []
```

```
for i in a
```

```
    if i in b and i not in common
```

```
        common.append(i)
```

```
print(common)
```

Question 46

Find duplicate elements in list

Input

[1, 2, 3, 2, 4, 1]

Output

[1, 2]

Answer

```
nums = [1, 2, 3, 2, 4, 1]
```

```
seen = []
```

```
dup = []
```

```
for n in nums
```

```
    if n in seen and n not in dup
```

```
        dup.append(n)
```

```
    else
```

```
        seen.append(n)
```

```
print(dup)
```

Question 47

Rotate list right by one

Input

[1, 2, 3, 4]

Output

[4, 1, 2, 3]

Answer

```
nums = [1, 2, 3, 4]
last = nums[-1]
result = [last] + nums[:-1]
print(result)
```

Question 48

Check if two lists are equal without using equals

Answer

```
a = [1, 2, 3]
b = [1, 2, 3]

is_same = True

if len(a) != len(b)
    is_same = False
else
    for i in range(len(a))
        if a[i] != b[i]
            is_same = False
            break

print(is_same)
```

Question 49

Merge two sorted lists

Input

[1, 3, 5] and [2, 4, 6]

Output

[1, 2, 3, 4, 5, 6]

Answer

a = [1, 3, 5]

b = [2, 4, 6]

result = []

i = 0

j = 0

while i < len(a) and j < len(b)

if a[i] < b[j]

result.append(a[i])

i += 1

else

result.append(b[j])

j += 1

result.extend(a[i:])

result.extend(b[j:])

print(result)

Question 50

Check balanced parentheses

Input

()()

Output

True

Answer

```
s = "()()"
count = 0
```

```
for ch in s
    if ch == "("
        count += 1
    else
        count -= 1
    if count < 0
        break
```

```
print(count == 0)
```

Question 51

Find pair with given sum

Input

[2, 4, 6, 8] sum 10

Output

(2, 8) (4, 6)

Answer

```
nums = [2, 4, 6, 8]
target = 10
```

```
for i in range(len(nums))
    for j in range(i + 1, len(nums))
        if nums[i] + nums[j] == target
            print(nums[i], nums[j])
```

Question 52

Reverse list without using reverse

Answer

```
nums = [1, 2, 3, 4]
```

```
rev = []
```

```
for i in range(len(nums) - 1, -1, -1)
```

```
    rev.append(nums[i])
```

```
print(rev)
```

Question 53

Check number is power of two

Input

16

Output

True

Answer

```
n = 16
```

```
if n > 0 and (n & (n - 1)) == 0
```

```
    print(True)
```

```
else
```

```
    print(False)
```

Question 54

Count frequency of elements in list

Input

[1, 2, 2, 3, 1]

Answer

```
nums = [1, 2, 2, 3, 1]
```

```
freq = {}
```

```
for n in nums
```

```
    freq[n] = freq.get(n, 0) + 1
```

```
print(freq)
```

Question 55

Find first repeating element

Input

[1, 2, 3, 2, 1]

Output

2

Answer

```
nums = [1, 2, 3, 2, 1]
```

```
seen = []
```

```
for n in nums
```

```
    if n in seen
```

```
        print(n)
```

```
        break
```

```
    seen.append(n)
```

Question 56

Check list is palindrome

Input

[1, 2, 3, 2, 1]

Output

True

Answer

```
nums = [1, 2, 3, 2, 1]
print(nums == nums[::-1])
```

Question 57

Find maximum sum subarray

Input

[1, -2, 3, 4, -1]

Output

7

Answer

```
nums = [1, -2, 3, 4, -1]
current = max_sum = nums[0]

for n in nums[1:]:
    current = max(n, current + n)
    max_sum = max(max_sum, current)

print(max_sum)
```

Question 58

Count occurrences of word in sentence

Input

ai gen ai agent ai

Word

ai

Output

3

Answer

```
s = "ai gen ai agent ai"
```

```
target = "ai"
```

```
count = 0
```

```
for w in s.split()
```

```
if w == target
```

```
count += 1
```

```
print(count)
```

Question 59

Find common characters in two strings

Input

genai and agent

Output

a g e n

Answer

```
s1 = "genai"
```



```
s2 = "agent"
common = ""

for ch in s1
    if ch in s2 and ch not in common
        common += ch

print(common)
```

Question 60

Convert decimal to binary

Input

10

Output

1010

Answer

```
n = 10
```

```
binary = ""
```

```
while n > 0
```

```
    binary = str(n % 2) + binary
```

```
    n = n // 2
```

```
print(binary)
```

Question 61

Check if string has all unique characters

Input

genai

Output

True

Answer

```
s = "genai"  
seen = ""  
unique = True
```

```
for ch in s  
    if ch in seen  
        unique = False  
        break  
    seen += ch  
  
print(unique)
```

Question 62

Find smallest number in list

Input

```
[5, 3, 9, 1, 6]
```

Output

1

Answer

```
nums = [5, 3, 9, 1, 6]  
smallest = nums[0]
```

```
for n in nums  
    if n < smallest  
        smallest = n
```

```
print(smallest)
```

Question 63

Check if two strings are rotation of each other

Input

abcd and cdab

Output

True

Answer

```
s1 = "abcd"
```

```
s2 = "cdab"
```

```
if len(s1) == len(s2) and s2 in s1 + s1
```

```
    print(True)
```

```
else
```

```
    print(False)
```

Question 64

Find common elements in three lists

Input

```
[1, 2, 3] [2, 3, 4] [3, 2, 5]
```

Output

```
[2, 3]
```

Answer

```
a = [1, 2, 3]
```

```
b = [2, 3, 4]
```

```
c = [3, 2, 5]
common = []

for n in a
    if n in b and n in c
        common.append(n)

print(common)
```

Question 65

Remove element at even index

Input

[10, 20, 30, 40, 50]

Output

[20, 40]

Answer

```
nums = [10, 20, 30, 40, 50]
result = []

for i in range(len(nums))
    if i % 2 != 0
        result.append(nums[i])

print(result)
```

Question 66

Find sum of prime numbers in list

Input

[2, 4, 5, 6, 7]

Output

14

Answer

```
nums = [2, 4, 5, 6, 7]
```

```
total = 0
```

```
for n in nums
```

```
    if n > 1
```

```
        prime = True
```

```
        for i in range(2, n)
```

```
            if n % i == 0
```

```
                prime = False
```

```
        break
```

```
    if prime
```

```
        total += n
```

```
print(total)
```

Question 67

Flatten nested list

Input

[1, [2, 3], [4, [5, 6]]]

Answer

```
def flatten(lst)
```

```
    result = []
```

```
    for item in lst
```

```
if type(item) == list
    result.extend(flatten(item))
else
    result.append(item)
return result

nums = [1, [2, 3], [4, [5, 6]]]
print(flatten(nums))
```

Question 68

Find longest consecutive sequence

Input

[100, 4, 200, 1, 3, 2]

Output

4

Answer

```
nums = [100, 4, 200, 1, 3, 2]
nums = list(set(nums))
nums.sort()
longest = 1
current = 1

for i in range(1, len(nums))
    if nums[i] == nums[i - 1] + 1
        current += 1
    longest = max(longest, current)
    else
        current = 1
```

```
print(longest)
```

Question 69

Check if number is perfect square

Input

16

Output

True

Answer

```
n = 16
```

```
i = 1
```

```
while i * i <= n
```

```
    if i * i == n
```

```
        print(True)
```

```
        break
```

```
    i += 1
```

```
else
```

```
    print(False)
```

Question 70

Count words starting with vowel

Input

ai is an autonomous agent

Output

3

Answer

```
s = "ai is an autonomous agent"  
count = 0
```

```
for w in s.split()  
    if w[0].lower() in "aeiou"  
        count += 1
```

```
print(count)
```

Question 71

Shift all negative numbers to left

Input

```
[1, -2, 3, -4, 5]
```

Output

```
[-2, -4, 1, 3, 5]
```

Answer

```
nums = [1, -2, 3, -4, 5]  
neg = []  
pos = []
```

```
for n in nums  
    if n < 0  
        neg.append(n)  
    else  
        pos.append(n)  
print(neg + pos)
```

Question 72

Check if string follows pattern

Input

abba

dog cat cat dog

Output

True

Answer

```
pattern = "abba"
```

```
words = "dog cat cat dog".split()
```

```
mapping = {}
```

```
used = {}
```

```
valid = True
```

```
for p, w in zip(pattern, words)
```

```
    if p in mapping
```

```
        if mapping[p] != w
```

```
            valid = False
```

```
            break
```

```
        else
```

```
            if w in used
```

```
                valid = False
```

```
                break
```

```
            mapping[p] = w
```

```
            used[w] = True
```

```
print(valid)
```

Question 73

Find equilibrium index

Input

[1, 3, 5, 2, 2]

Output

2

Answer

```
nums = [1, 3, 5, 2, 2]
```

```
total = sum(nums)
```

```
left = 0
```

```
for i in range(len(nums))
```

```
    total -= nums[i]
```

```
    if left == total
```

```
        print(i)
```

```
        break
```

```
    left += nums[i]
```

Question 74

Find majority element

Input

[2, 2, 1, 2, 3, 2, 2]

Output

2

Answer

```
nums = [2, 2, 1, 2, 3, 2, 2]
```

```
count = 0
candidate = None

for n in nums:
    if count == 0:
        candidate = n
    count += 1 if n == candidate else -1

print(candidate)
```

Question 75

Check if matrix is identity matrix

Answer

```
matrix = [[1,0,0],[0,1,0],[0,0,1]]
identity = True

for i in range(len(matrix)):
    for j in range(len(matrix)):
        if i == j and matrix[i][j] != 1:
            identity = False
        if i != j and matrix[i][j] != 0:
            identity = False

print(identity)
```

Question 76

Rotate matrix clockwise

Input

```
[[1,2],[3,4]]
```

Output

```
[[3,1],[4,2]]
```

Answer

```
matrix = [[1,2],[3,4]]
```

```
n = len(matrix)
```

```
rotated = []
```

```
for i in range(n)
```

```
    row = []
```

```
    for j in range(n)
```

```
        row.append(matrix[n - j - 1][i])
```

```
    rotated.append(row)
```

```
print(rotated)
```

Question 77

Find gcd of two numbers

Input

24 and 36

Output

12

Answer

```
a = 24
```

```
b = 36
```

```
while b != 0
```

```
    a, b = b, a % b
```

```
print(a)
```

Question 78

Check if string can be rearranged to palindrome

Input

carrace

Output

True

Answer

```
s = "carrace"
```

```
freq = {}
```

```
odd = 0
```

```
for ch in s
```

```
    freq[ch] = freq.get(ch, 0) + 1
```

```
for v in freq.values()
```

```
    if v % 2 != 0
```

```
        odd += 1
```

```
print(odd <= 1)
```

Question 79

Find missing characters in string

Input

abcdef and acdf

Output

b e

Answer

```
s1 = "abcdef"
```

```
s2 = "acdf"
```

```
missing = ""
```

```
for ch in s1
```

```
    if ch not in s2
```

```
        missing += ch + " "
```

```
print(missing)
```

Question 80

Check if number is strong number

Input

145

Output

True

Answer

```
n = 145
```

```
temp = n
```

```
total = 0
```

```
while temp > 0
```

```
    digit = temp % 10
```

```
    fact = 1
```

```
    for i in range(1, digit + 1)
```

```
        fact *= i
```

```
    total += fact
```

```
    temp = temp // 10
```

```
print(total == n)
```

Question 81

Reverse number using logic

Input

1234

Output

4321

Answer

```
n = 1234
```

```
rev = 0
```

```
while n > 0
```

```
    rev = rev * 10 + n % 10
```

```
    n = n // 10
```

```
print(rev)
```

Question 82

Check if number is even or odd without modulo

Input

7

Output

Odd

Answer

```
n = 7
```

```
if (n & 1) == 0
    print("Even")
else
    print("Odd")
```

Question 83

Find sum of first n natural numbers

Input

10

Output

55

Answer

n = 10

total = n * (n + 1) // 2

print(total)

Question 84

Count uppercase and lowercase letters

Input

GeNai

Output

Uppercase 2 Lowercase 3

Answer

s = "GeNai"


```
upper = 0
lower = 0

for ch in s:
    if ch.isupper():
        upper += 1
    elif ch.islower():
        lower += 1

print(upper, lower)
```

Question 85

Remove vowels from string

Input

generative

Output

gnrtv

Answer

```
s = "generative"
result = ""
```

```
for ch in s:
    if ch not in "aeiou":
        result += ch

print(result)
```

Question 86

Find sum of elements at even index

Input

[10, 20, 30, 40]

Output

40

Answer

```
nums = [10, 20, 30, 40]
```

```
total = 0
```

```
for i in range(len(nums))
```

```
    if i % 2 == 0
```

```
        total += nums[i]
```

```
print(total)
```

Question 87

Check if list contains duplicate

Input

[1, 2, 3, 1]

Output

True

Answer

```
nums = [1, 2, 3, 1]
```

```
seen = []
```

```
duplicate = False
```

```
for n in nums
```

```
    if n in seen
```

```
        duplicate = True
```

```
break
seen.append(n)

print(duplicate)
```

Question 88

Find sum of digits until single digit

Input

9875

Output

2

Answer

```
n = 9875
```

```
while n > 9
```

```
total = 0
```

```
while n > 0
```

```
total += n % 10
```

```
n = n // 10
```

```
n = total
```

```
print(n)
```

Question 89

Check if number is Harshad number

Input

18

Output

True

Answer

n = 18

temp = n

total = 0

while temp > 0

total += temp % 10

temp = temp // 10

print(n % total == 0)

Question 90

Count trailing zeros in factorial

Input

5

Output

1

Answer

n = 5

count = 0

while n > 0

n = n // 5

count += n

print(count)

Question 91

Find product of all elements in list

Input

[1, 2, 3, 4]

Output

24

Answer

```
nums = [1, 2, 3, 4]
```

```
product = 1
```

```
for n in nums
```

```
    product *= n
```

```
print(product)
```

Question 92

Check if string contains only alphabets

Input

GenAI

Output

True

Answer

```
s = "GenAI"
```

```
print(s.isalpha())
```

Question 93

Find minimum difference between elements

Input

[1, 5, 9, 15]

Output

4

Answer

```
nums = [1, 5, 9, 15]
nums.sort()
min_diff = nums[1] - nums[0]

for i in range(1, len(nums) - 1):
    diff = nums[i + 1] - nums[i]
    if diff < min_diff:
        min_diff = diff

print(min_diff)
```

Question 94

Print pattern

1

12

123

Answer

n = 3

```
for i in range(1, n + 1):
    for j in range(1, i + 1):
```

```
print(j, end="")  
print()
```

Question 95

Check if year is leap year

Input

2024

Output

True

Answer

```
year = 2024
```

```
if year % 400 == 0 or (year % 4 == 0 and year % 100 != 0)
```

```
    print(True)
```

```
else
```

```
    print(False)
```

Question 96

Find average of list elements

Input

[10, 20, 30]

Output

20

Answer

```
nums = [10, 20, 30]
```

```
total = 0
```

```
for n in nums
    total += n

print(total / len(nums))
```

Question 97

Replace spaces with underscore

Input

gen ai agent

Output

gen_ai_agent

Answer

```
s = "gen ai agent"
result = ""
```

```
for ch in s
    if ch == " "
        result += "_"
    else
        result += ch

print(result)
```

Question 98

Check if two numbers are coprime

Input

8 and 15

Output

True

Answer

a = 8

b = 15

while b != 0

a, b = b, a % b

print(a == 1)

Question 99

Find cube of each element in list

Input

[1, 2, 3]

Output

[1, 8, 27]

Answer

nums = [1, 2, 3]

cubes = []

for n in nums

cubes.append(n ** 3)

print(cubes)

Question 100

Check if string is pangram

Input

the quick brown fox jumps over the lazy dog

Output

True

Answer

```
s = "the quick brown fox jumps over the lazy dog"
```

```
alphabet = "abcdefghijklmnopqrstuvwxyz"
```

```
is_pangram = True
```

```
for ch in alphabet
```

```
    if ch not in s
```

```
        is_pangram = False
```

```
        break
```

```
print(is_pangram)
```